

Dimensionality Reduction

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Contents

1	Introduction	2
2	Principal Component Analysis	2
2.1	Introduction	2
2.2	Derivation of PCA	2
2.2.1	Maximum Variance Formulation	2
2.2.2	Minimum Error Formulation	2
2.3	Algorithm	2
2.4	Application of PCA	3
2.5	Extensions of PCA	3
2.5.1	Probabilistic PCA	3
2.5.2	Kernel PCA	4
3	Linear Discriminant Analysis	4

1 Introduction

This document presents the methods of **dimensionality reduction**, which embeds dataset in the higher dimension to a lower dimension

2 Principal Component Analysis

2.1 Introduction

Principal component analysis (PCA) is the most simple dimensionality reduction method. PCA apply a linear projection from high dimension to a lower dimension.

2.2 Derivation of PCA

We can derive PCA in two ways; by maximizing the variance of transformed dataset, or by minimizing the projection error. The main idea is that we assume that the dataset can be formed as a linear combination of lower dimensional embedding vectors, which are believed to be managable then the original dataset, i.e., $\mathbf{x} = \mathbf{W}\mathbf{z}$. And our objective is to find out the matrix $\mathbf{W}$ that best

2.2.1 Maximum Variance Formulation

TODO

2.2.2 Minimum Error Formulation

TODO

2.3 Algorithm

Algorithm 1: Principal component analysis

Input : dataset $\mathbf{X} \in \mathbb{R}^{N \times D}$

lower dimension $M < D$

Output: dataset $\tilde{\mathbf{X}} \in \mathbb{R}^{N \times M}$ with reduced dimension

- 1 $\bar{\mathbf{x}} \leftarrow \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n$
 - 2 $\mathbf{S} \leftarrow \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \bar{\mathbf{x}})(\mathbf{x}_n - \bar{\mathbf{x}})^\top = \frac{1}{N} \bar{\mathbf{X}}^\top \bar{\mathbf{X}}$
 - 3 $\{\mathbf{u}_i\}_{i \in \{1, \dots, M\}} \leftarrow$ normalized eigenvectors of $\mathbf{S}$ with associated eigenvalues in decreasing order
 - 4 $\mathbf{U} \leftarrow \begin{bmatrix} \mathbf{u}_1^\top \\ \vdots \\ \mathbf{u}_M^\top \end{bmatrix}$
 - 5 $\tilde{\mathbf{X}} \leftarrow \mathbf{X}\mathbf{U}^\top$
 - 6 **return** $\tilde{\mathbf{X}}$
-

The matrix $\mathbf{S}$ is called **data covariance matrix**.

When the dimensionality of the data is substantially greater than the number of data, i.e., $N \ll D$, the original PCA is infeasible, since its computational cost is $\mathcal{O}(D^3)$. Also, since N points defines a linear subspace whose dimensionality is at most $N - 1$, there is little point in applying PCA for dimension higher than N . We can overcome this with a slight modification of the original algorithm, with computational cost $\mathcal{O}(N^3)$.

Algorithm 2: Modified PCA

Input : dataset $\mathbf{X} \in \mathbb{R}^{N \times D}$

lower dimension $M < N$

Output: dataset $\tilde{\mathbf{X}} \in \mathbb{R}^{N \times M}$ with reduced dimension

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1  $\bar{\mathbf{x}} \leftarrow \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n$ 
2  $\bar{\mathbf{X}} \leftarrow \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_N^\top \end{bmatrix}$ 
3  $\mathbf{S}' \leftarrow \frac{1}{N} \bar{\mathbf{X}} \bar{\mathbf{X}}^\top$ 
4  $\{\mathbf{v}_i\}_{i \in \{1, \dots, M\}} \leftarrow$  eigenvectors of  $\mathbf{S}'$  with associated eigenvalues in decreasing order
5  $\mathbf{u}_i \leftarrow \mathbf{X}^\top \mathbf{v}_i$ 
6  $\mathbf{u}_i \leftarrow \frac{\mathbf{u}_i}{\|\mathbf{u}_i\|}$ 
7  $\mathbf{U} \leftarrow \begin{bmatrix} \mathbf{u}_1^\top \\ \vdots \\ \mathbf{u}_M^\top \end{bmatrix}$ 
8  $\tilde{\mathbf{X}} \leftarrow \mathbf{X} \mathbf{U}^\top$ 
9 return  $\tilde{\mathbf{X}}$ 

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2.4 Application of PCA

PCA is mainly used for data compression.

PCA can be used in data preprocessing, which is called *whitening* or *sphering* the data. It does the same thing what *standard scaling* does, but with a more substantial normalization, making different variables decorrelated. To do this, we first write the eigenvector equation in Algorithm 1 as

$$\mathbf{S} \mathbf{U}^\top = \mathbf{U}^\top \mathbf{L}$$

where $\mathbf{S}$ is data covariance matrix, and $\mathbf{L} = \text{diag}(\lambda_1, \dots, \lambda_N)$. Then we transform each data vector as

$$\mathbf{y}_n = \mathbf{L}^{-1/2} \mathbf{U} (\mathbf{x}_n - \bar{\mathbf{x}}).$$

The set $\{\mathbf{y}_n\}$ has zero mean and identity covariance.

2.5 Extensions of PCA

2.5.1 Probabilistic PCA

TODO

2.5.2 Kernel PCA

TODO

3 Linear Discriminant Analysis

TODO